Numericals
Chap # 8

Q1

Data:-

a) $F = 0.1N$

$r = 50cm = 0.5m$

$q_1 = q_2 = ?$

b) $\epsilon_r = 10$

$q_1 = q_2 = ?$

Sol:-

A/c to coulombs law:-

$$F = \frac{kq_1q_2}{r^2}$$

$$0.1 = \frac{9\times10^9 \cdot q \cdot q}{(0.5)^2}$$

$$\frac{0.1\times0.25}{9\times10^9} = q^2$$

Teacher's Signature

$q^2 = 2.77 \times 10^{-12}$

$q = \sqrt{2.77 \times 10^{-12}}$

$q = 1.66 \times 10^{-6}\ C$

$q = 1.66\ \mu C$

b) If $\varepsilon_r = 10$

$$F = K \varepsilon_r \times \frac{q \cdot q}{r^2}$$

$$0.1 = 9 \times 10^9 \times 10 \times \frac{q^2}{(0.5)^2}$$

$$\frac{0.025}{9 \times 10^{10}} = q^2$$

$q = 5.3\ \mu C$

Q2

Data:-

$m_p = 1.67 \times 10^{-27}\ kg$

$q = 1.6 \times 10^{-19}\ C$

$r = ?$

Sol:-

A/c to given condition

Date

electrostatic force = gravitational force

$$\frac{kq_1q_2}{r^2} = mg$$

$$\Rightarrow r^2 = \frac{kq_1q_2}{mg}$$

$$r^2 = \frac{9\times10^9 \times 1.6\times10^{-19}\times1.6\times10^{-19}}{1.67\times10^{-27}\times9.8}$$

$$r = 0.119\, m$$

Q3

$q = 1.6\times10^{-19}\, C$

$E = 1200\, v/cm = \frac{1200}{1/100} = 120000\, v/m$

$r = 2cm = 0.02m$

$m_e = 9.1\times10^{-31}\, kg$

a) $F = ?$

b) $a = ?$

c) $t = ?$

Sol:-

a) $F = Eq$

$F = 120000\times 1.6\times10^{-19}$

$F = 1.92\times10^{-14}$

b) $a = \frac{F}{m} = \frac{1.92 \times 10^{-4}}{9.1 \times 10^{-31}}$

$a = 2.1 \times 10^{16}\ m/sec^2$

c) $S = v_i t + \frac{1}{2} a t^2$

$S = 0 + \frac{1}{2}(2.1 \times 10^{16}) t^2$

$\frac{0.02 \times 2}{2.1 \times 10^{16}} = t^2$

$t = \sqrt{\frac{0.02 \times 2}{2.1 \times 10^{16}}}$

$t = 1.37 \times 10^{-9} sec$

Data:-

mass $= 6.64 \times 10^{-27} kg$

$q = 2e^- = 2 \times 1.6 \times 10^{-19} C = 3.2 \times 10^{-19} C$

$E = ?$

Sol:-

A/c to given condition:-

Electrostatic force = Gravitational force

$F = w$

$Eq = mg$

Date

$$E = \frac{mg}{q}$$

$$E = \frac{(6.64\times10^{-27})(9.8)}{(3.2\times10^{-19})}$$

$$E = 2.03\times10^{-7}\ N/C$$

electric field is directed upward.

Q5

Data:

Length of side = 2×10^{-6} m

$q_p = q_e = 1.6\times10^{-19}$ C

Net intensity at third corner = E = ?

Sol:-

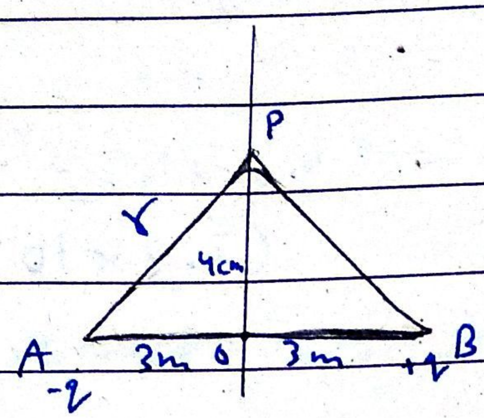
$$E = E(+\cos\theta) + E(-\cos\theta)$$

$$E = \frac{kq}{r^2}\cos\theta + \frac{kq}{r^2}\cos\theta$$

$$E = 2\left(\frac{kq}{r^2}\right)\cos\theta$$

$$E = 2\left[\frac{9\times10^{9}\times1.6\times10^{-19}\times\cos60^\circ}{(2\times10^{-6})^2}\right]$$

$$E = 360\ N/C$$



Data:-

Electric intensity at P = ?

$-q = 3.20 \times 10^{-19}\,C$

$+q = 3.20 \times 10^{-19}\,C$

Sol:-

In ΔAOP

$$|AP|^2 = |AO|^2 + |OP|^2$$

$$r^2 = 3^2 + 4^2$$

$$r = \sqrt{9+16}$$

$$r = \sqrt{25}$$

$$r = 5m$$

$$\cos\theta = \frac{AO}{AP}$$

$$\cos\theta = \frac{3}{5}$$

$$\boxed{\cos\theta = 0.6}$$

$\therefore E = E_+ \cos\theta + E_- \cos\theta$

$E = \frac{kq}{r^2}\cos\theta + \frac{kq}{r^2}\cos\theta$

$E = \frac{2kq}{r^2}\cos\theta$

$E = \frac{2 \times 9 \times 10^9 \times 3.2 \times 10^{-19} \times 0.6}{5^2}$

direction along $-x$ axis

Q7

Data:

$q = 1.6 \times 10^{-19}\ C$

$r = 5\ \mu m = 5 \times 10^{-6}\ m$

$E = ?$

Sol:

$E = E_+ \cos\theta + E_- \cos\theta$

$E = \frac{2kq}{r^2}\cos\theta$

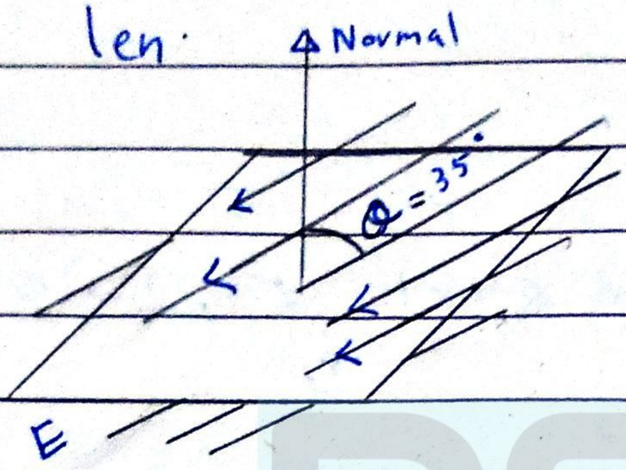
$E = \frac{2 \times 9 \times 10^9 \times 1.9 \times 10^{-6} \times \cos 60}{(5 \times 10^6)^2}$

$E = 57.6\ N/C$

Q8

Data:-

len.



Data:-

length of each side = 3.2mm = 3.2×10^{-3}m

$E = 1800$ N/C

$\theta = 35^\circ$ with normal

electric flux = ϕ = ?

Sol:-

Angle b/w E and ΔA

$\theta' = 180 - 35$

$\theta' = 145$

$\phi = \vec{E} \cdot \Delta\vec{A}$

$\phi = E \Delta A \cos\theta$

$\phi = 1800 \times l^2 \times \cos 145^\circ$

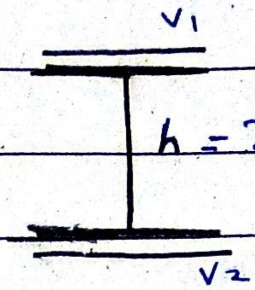
$\phi = 1800 \times (3.2 \times 10^{-3})^2 \times (-0.8191)$

$\phi = 1.5 \times 10^{-2}$ Nm²/C

Date

Q9

Data:-



$h = 2cm = 0.02m$

$V_1 = 2400V$

$V_2 = 0V$

$t = ?$

Sol:-

$\therefore F = Eq$

$$F = \frac{V}{d} q$$

$$F = \frac{2400}{0.02} \times 1.6 \times 10^{-19}$$

$$F = 1.92 \times 10^{-14} N$$

$$\therefore a = \frac{f}{m}$$

$$a = \frac{1.92 \times 10^{-14}}{9.1 \times 10^{-31}}$$

$$a = 2.11 \times 10^{16} m/s^2$$

$$\therefore S = V_i t + \frac{1}{2} a t^2$$

Date

This pdf might have some mistakes

$$t^2 = \frac{2s}{a}$$

$$t = \sqrt{\frac{2(0.02)}{2.11\times10^{16}}}$$

$t = 1.37\times10^{-9}$ sec

$t = 1.37$ nsec

Q10

Data:

$d = 1.5\text{cm} = 0.015\text{cm}$

$V_+ = 5V$

$V_- = 0V$

$\Delta V = 5-0 = 5V$

$E = ?$

Sol:

Halfway b/w the plates

$$d = \frac{0.015}{2} = \mathbf{0.0075}$$

$$E = \frac{V}{d} = \frac{5}{0.0075}$$

$E = 666.66$ N/C